

GNN-Based Network Intrusion Detection

Evaluation Results — E-GraphSAGE vs. HeteroGNN — UNSW-NB15 and CICIDS Test Sets

Summary — Best Runs Across Datasets

Dataset	Model	Best Val F1	Test Macro-F1	Weighted-F1	MCC	Kappa	Train Time	Epoch
UNSW	E-GraphSAGE	0.5097	0.5022	0.9815	0.8279	0.8276	9.2 min	129
	HeteroGNN	0.4743	0.4882	0.9814	0.8398	0.8397	8.7 min	90
CICIDS	E-GraphSAGE	0.6965	0.6380	0.9835	0.9309	0.9305	79.5 min	151
	HeteroGNN	0.7004	0.6203	0.9840	0.9350	0.9344	39.3 min	60

Accuracy is reported per model inside each dataset section but is omitted here since it is misleading under severe class imbalance; Macro-F1 and per-class F1 are the primary comparison metrics. Two model architectures — E-GraphSAGE (homogeneous edge classification) and HeteroGNN (heterogeneous graph formulation) — were evaluated on each dataset.

UNSW — UNSW-NB15

E-GraphSAGE (early stopping at epoch 129)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9819	0.5022	0.9815	0.8279	0.8276
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9539	0.9990	0.4758	0.5097	9.2 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.2570	0.9892	0.4080	185	0.9922	0.2279
Backdoor	0.0067	0.0043	0.0052	700	0.6790	0.0169
Benign	1.0000	0.9998	0.9999	333,440	1.0000	1.0000
DoS	0.3613	0.1918	0.2505	897	0.9843	0.2376
Exploits	0.7729	0.6379	0.6990	6,413	0.9943	0.7663
Fuzzers	0.6407	0.9074	0.7511	5,073	0.9911	0.6573
Generic	0.7972	0.7664	0.7815	2,949	0.9953	0.8123
Reconnaissance	0.8584	0.5086	0.6387	2,562	0.9935	0.7021
Shellcode	0.2986	0.6173	0.4026	358	0.9180	0.3105
Worms	0.0909	0.0800	0.0851	25	0.9918	0.0275

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
1,918	1,671	71.95%	0.5363	2.9495

HeteroGNN (early stopping at epoch 90)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9832	0.4882	0.9814	0.8398	0.8397
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.8183	0.9979	0.5104	0.4743	8.7 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Analysis	0.8909	0.2649	0.4083	185	0.9901	0.5281
Backdoor	0.0000	0.0000	0.0000	700	0.1881	0.0013
Benign	0.9998	1.0000	0.9999	333,440	0.9996	0.9999
DoS	0.6154	0.0089	0.0176	897	0.9747	0.2069
Exploits	0.6839	0.7561	0.7182	6,413	0.9924	0.7817
Fuzzers	0.6418	0.8573	0.7341	5,073	0.9952	0.6417
Generic	0.7597	0.8094	0.7838	2,949	0.9923	0.7567
Reconnaissance	0.8239	0.5550	0.6632	2,562	0.9941	0.7121
Shellcode	0.6367	0.4944	0.5566	358	0.8086	0.4621
Worms	0.0000	0.0000	0.0000	25	0.2473	0.0134

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
1,703	6,684	100.00%	0.6609	2.4714

CICIDS — CIC-IDS2018

E-GraphSAGE (early stopping at epoch 151)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9864	0.6380	0.9835	0.9309	0.9305
ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9840	0.9994	0.6568	0.6965	79.5 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Benign	0.9931	0.9980	0.9956	2,608,914	0.9842	0.9976
Bot	0.9999	0.9977	0.9988	16,719	0.9996	0.9977
Brute_Force_-Web	0.1554	0.5164	0.2389	244	0.9985	0.1135
Brute_Force_-XSS	0.0000	0.0000	0.0000	69	0.9999	0.2350
DDOS_attack-HOIC	0.9999	1.0000	1.0000	154,848	1.0000	1.0000
DDOS_attack-LOIC-UDP	0.2719	0.9423	0.4220	260	0.9576	0.2234
DDoS_attacks-LOIC-HTTP	1.0000	0.9487	0.9737	43,289	0.9853	0.9847
DoS_attacks-GoldenEye	1.0000	0.7747	0.8730	4,598	0.9997	0.8914
DoS_attacks-Hulk	1.0000	0.8372	0.9114	15,012	1.0000	0.9995
DoS_attacks-SlowHTTPTest	0.0000	0.0000	0.0000	15,833	0.9958	0.4329
DoS_attacks-Slowloris	1.0000	0.8302	0.9072	2,703	0.9999	0.9576
FTP-BruteForce	0.6469	1.0000	0.7856	29,004	0.9977	0.6854
Infiltration	0.5630	0.2811	0.3750	17,372	0.8418	0.2950
SQL_Injection	0.0477	0.6970	0.0893	66	0.9993	0.0376
SSH-Bruteforce	1.0000	1.0000	1.0000	14,137	1.0000	1.0000

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
3,389	3,051	93.69%	0.1686	1.4938

HeteroGNN (early stopping at epoch 60)

Accuracy	Macro-F1	Weighted-F1	MCC	Cohen's Kappa
0.9872	0.6203	0.9840	0.9350	0.9344

ROC-AUC (macro)	ROC-AUC (micro)	PR-AUC (macro)	Best Val F1	Training Time
0.9275	0.9996	0.6675	0.7004	39.3 min

Per-Class Performance

Class	Prec	Recall	F1	Support	ROC-AUC	PR-AUC
Benign	0.9929	0.9991	0.9960	2,608,914	0.9958	0.9993
Bot	0.9997	0.9566	0.9777	16,719	1.0000	0.9992
Brute_Force_-Web	0.6250	0.0205	0.0397	244	0.8126	0.3793
Brute_Force_-XSS	0.0000	0.0000	0.0000	69	0.4087	0.0174
DDOS_attack-HOIC	1.0000	1.0000	1.0000	154,848	1.0000	1.0000
DDOS_attack-LOIC-UDP	0.2554	0.9038	0.3983	260	0.9496	0.2065
DDoS_attacks-LOIC-HTTP	1.0000	0.9487	0.9737	43,289	0.9855	0.9824
DoS_attacks-GoldenEye	0.9955	0.7749	0.8715	4,598	1.0000	0.9941
DoS_attacks-Hulk	1.0000	0.8372	0.9114	15,012	1.0000	1.0000
DoS_attacks-SlowHTTPTest	0.0000	0.0000	0.0000	15,833	0.9950	0.3531
DoS_attacks-Slowloris	1.0000	0.8657	0.9280	2,703	0.9998	0.9364
FTP-BruteForce	0.6469	1.0000	0.7856	29,004	0.9973	0.6469
Infiltration	0.6996	0.3025	0.4224	17,372	0.9455	0.4202
SQL_Injection	0.0000	0.0000	0.0000	66	0.8233	0.0777
SSH-Bruteforce	1.0000	1.0000	1.0000	14,137	1.0000	1.0000

GNN Structural Metrics

Avg. Nodes/Graph	Avg. Edges/Graph	Neighbor Coverage	Silhouette	Davies-Bouldin
3,566	12,205	100.00%	0.3800	1.2362